

WASP-36b

R-1476

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_220207 · created 2026-08-01T12:23:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459617.7212 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.048
Transit depth (Rp/Rs)^2	2.73 +/- 1.58 [%]
Orbital Inclination (inc)	87.99 +/- 2.88
Airmass coefficient 1 (a1)	0.59 +/- 0.3
Airmass coefficient 2 (a2)	0.25 +/- 0.41
Scatter in the residuals of the lightcurve fit is	48.73 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	6.75
Transit Duration (day)	0.0933 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.048 vs 0.1368 (deviation 0.48σ)
Duration fit vs published	0.0933 d vs 0.0758 d (deviation 1.85σ)
Mid-time O–C	-0.5 min
Depth SNR	2.8
Residual scatter	48.73 %

Light curve

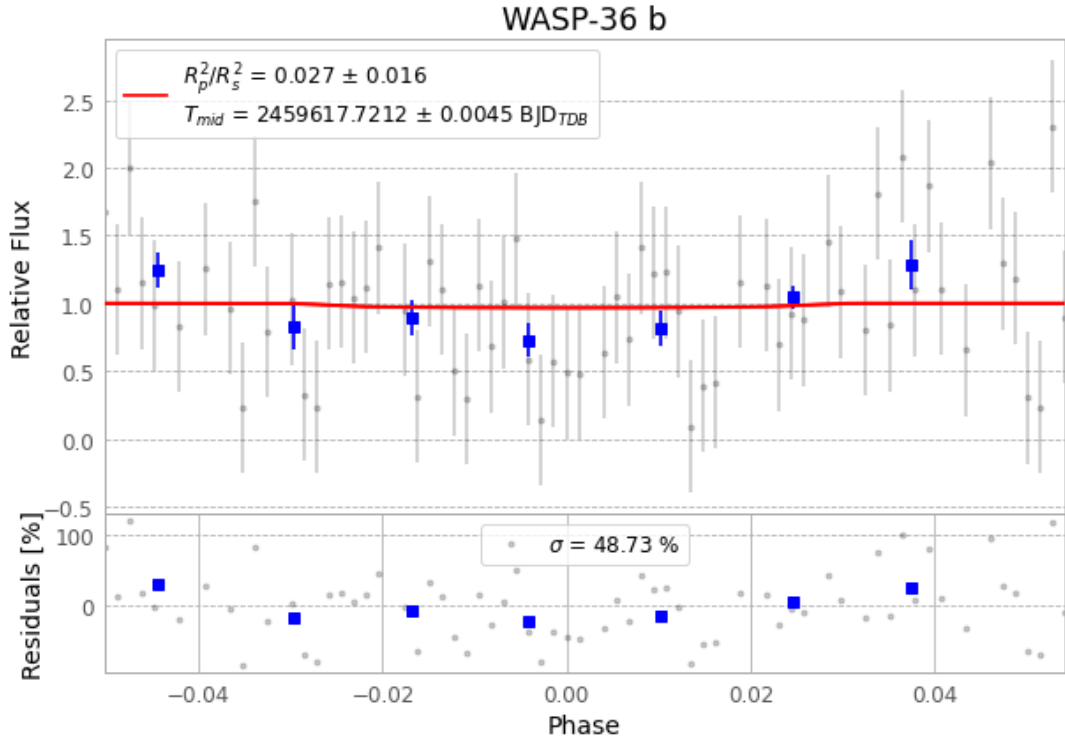


Figure 1 — FinalLightCurve_WASP-36 b_2022-02-06.png (SHA-256 d93752b2af06d9fa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [458, 455], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 526fc1ae2efab441...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

78 FITS frames (51 MB), WASP-36220207031812.FITS → WASP-36220207070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-220207.log (2bca4d479ad2...), FinalLightCurve_WASP-36 b_2022-02-06.png (d93752b2af06...), FinalLightCurve_WASP-36 b_2022-02-06.csv (31d7dce6b188...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 12:23Z.